

weights per feature
(n+1)

BIAS	x1	x2
1	1	2
1	3	4
1	5	6

X (m, n+1)
BIAS

SWITCH
row & col

X.T

BIAS	1	1	1
x1	1	3	5
x2	2	4	6

X (n+1, m)

1) $z = X \hat{a} w$ dot product,
(m x n+1) • (n+1)
applies weights accordingly
+ sums rows

$z.shape() = (n \times 1)$

$$\begin{aligned} 1 \cdot 0.5 + 1 \cdot 0.1 + 2 \cdot 0.2 &= 1.0 \\ 1 \cdot 0.5 + 3 \cdot 0.1 + 4 \cdot 0.2 &= 1.6 \\ 1 \cdot 0.5 + 5 \cdot 0.1 + 6 \cdot 0.2 &= 2.2 \end{aligned}$$

2) $\text{sigmoid} = 1 / (1 + \text{np.exp}(-z))$
the 'y hat per row'

3) $(1/m) \cdot X.T \cdot \text{np.dot}(\text{sigmoid} - y_{\text{true}})$

get
average
errors

error per row
applies errors feature wise

$$\begin{aligned} 1 \cdot 0.731 + 1 \cdot (-0.168) + 1 \cdot (-0.100) &= 0.463 \\ 1 \cdot 0.731 + 3 \cdot (-0.168) + 5 \cdot (-0.100) &= -0.273 \\ 2 \cdot 0.31 + 4 \cdot (-0.168) + 6 \cdot (-0.100) &= 0.190 \end{aligned}$$

GRADIENT,

measures loss sensitivity to each weight